Predictive State-Estimation MPPT using a Kalman Filter for Enhanced Dynamic Performance and Stability of a PV-Fed DC Motor Drive in an LVDC Microgrid

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***Abstract*—The increasing integration of photovoltaic (PV) systems into Low Voltage DC (LVDC) microgrids presents significant challenges in maintaining power stability, particularly when supplying dynamic loads. This paper presents a rigorous comparative analysis of two Maximum Power Point Tracking (MPPT) strategies for a PV-fed Separately Excited DC Machine (SEM) within an LVDC microgrid architecture. The study contrasts the performance of the conventional Perturb & Observe (P&O) algorithm augmented with a Proportional-Integral (PI) controller against a proposed predictive control strategy based on a Kalman filter. The Kalman filter approach reframes MPPT as a state-estimation problem, enabling it to effectively mitigate the noise measurement and predict the system's optimal operating point under fluctuating conditions. Through detailed simulations in MATLAB/Simulink under severe irradiance transients, the performance of both controllers is evaluated against key metrics. The results demonstrate that the Kalman filter-based MPPT achieves a tracking efficiency of 98.5% with virtually zero steady-state power ripple, a stark improvement over the P&O-PI controller's 92% efficiency and noticeable power oscillations. This enhanced electrical stability translates directly to superior mechanical performance, with the Kalman filter reducing the SEM's speed ripple to just ±0.2%, compared to the ±1.5% ripple induced by the P&O-PI controller. The findings conclusively establish that the predictive state-estimation approach offers significant advantages in tracking speed, stability, and overall energy yield, making it a critical enabling technology for reliable, high-performance LVDC microgrids.**

***Keywords*—Kalman filter, Low voltage DC (LVDC), Microgrid, MPPT, P&O Algorithm, PI controller, Separately Excited DC Machine (SEM).**

I. INTRODUCTION

A Inherent transformation is reshaping the global energy landscape by the two-fold drivers of decarbonization and decentralization. Against the backdrop of decarbonization and decentralization, Low Voltage DC (LVDC) microgrids now represent a highly compelling architecture for future power distribution. LVDC systems exhibit several natural advantages in comparison to traditional alternating current (AC) systems including improved end-to-end efficiency attributed to fewer stages of power conversion, the direct compatibility with the renewable energy sources (RES) that are DC-native including photovoltaics (PV) and energy storage systems (ESS), and the simple emission of considerations around reactive power and frequency stabilization [3]. Together, these advantages mean LVDC microgrids are better suited for integrating the increasing number of electronic loads associated with modern residential and commercial applications.

Despite these advantages, the widespread use of LVDC microgrids depends on severe technical difficulties. The high penetration of intermittent RES, i.e. solar, can bring variability to the system that may prevent stable operation and voltage regulation. The active microgrid adopts an active structure which allows for bidirectional flow, whereas passive networks have a passive structure and only allow for unidirectional flow. The nature of the fault current will be changed in significant ways and will require enhanced protection schemes that are generally more sophisticated and require quicker reaction time. In addition, the control and energy management of various distributed energy resources, distributed storage units, and dynamic loads must also be coordinated and stable to deliver dependable power quality and reliability. For example, performing an operation from the conventional grid-connected operational mode to islanding requires special on/off operations to assure sustainable use of the distributed energy resources.

A. The Critical role and Limitation of Conventional MPPT

When harvesting energy from a solar array integrated into a PV-connected LVDC microgrid, the priority is to maximize energy harvesting. The output power provided by the PV panel is dependent on time-dependent ambient conditions, namely solar irradiance and temperature. This necessitates a Maximum Power Point Tracking (MPPT) algorithm to ensure that the panel remains working at the point of maximum efficiency. Of the MPPT algorithms implemented, Perturb & Observe (P&O) is most often used in commercial designs, due to its performance, simplicity and low-cost installation [1].

Despite the attractive features of simplicity and performance associated with P&O, there are several serious operational drawbacks, especially when considering the dynamic aspects of ambient conditions [5]. Firstly, there is a fundamental limitation with P&O, which is the inherent tradeoff between speed of tracking and accuracy of steady-state tracking, which is determined by the fixed perturbation step. Accordingly, a large perturbation step achieves a fast response to ambient conditions, but higher power oscillation around the Maximum Power Point (MPP) leads to energy being dissipated as waste energy on a continuous basis. A small perturbation step size ensures oscillation is low, but system response to changes in ambient conditions then becomes slower, with the system slow to direct power harvesting towards changes.

Additionally, during times of swift changes in irradiance, for example by clouds that move quickly, the P&O algorithm may misinterpret the change in power and diverge away from the MPP and could lose substantial tracking. Moreover, under non-uniform irradiance, or partial shading conditions, the P-V curve of the PV array will have several local maxima. Due to the simple hill-climbing nature of the P&O algorithm it cannot differentiate between local and global peaks and will easily become stuck at a local MPPT which is not optimum and could lead to potential degradation in the system energy yield. These points highlight a need for intelligent MPPT techniques that can work successfully under the full range of actual working conditions.

B. State-of-the-Art and the Proposed Contribution

To mitigate the drawbacks associated with traditional MPPT there are numerous advanced strategies designed, including techniques available based on Fuzzy Logic Control (FLC) and Artificial Neural Networks (ANN). While these strategies can provide better results, they usually bring added complexity to designing and tuning and have higher computational requirements, as additions to the system proposed. One alternative and very powerful approach is to consider the MPPT problem not as a blind search problem, but rather as a state-estimation problem. This approach is advantageous in systems that have a lot of noise and variation, since state-estimation methods use a mathematical model of the system to condition noisy measurements and obtain an optimal estimate of the true state of the system.

The Kalman filter is a recursively applied, optimal state-estimation algorithm that is extremely suitable for this application [9]. It could predict the future state of the system and corrects the prediction with real-time measurements; it is resilient to both sensor noise and disturbances on the system. The predictive nature of the Kalman filter makes it more stable and quicker to converge than mainstream reactive techniques, such as P&O.

The major achievement of this paper is the design, implementation, and comparative validation of a Kalman filter-based predictive MPPT algorithm for a LVDC microgrid with a significantly dynamic load - a Separately Excited DC Machine (SEM). The use of Kalman filters for MPPT control has previously been suggested, but this study is unique in establishing the causal dependencies between the MPPT control effectiveness and the electromechanical stability of the end-load [9]. Through supplementary simulation of power oscillations experienced by the P&O algorithm, we have presented clear evidence that they were not only regrettable electrical losses, but primary mechanical speed ripples, and torque ripples in the DC motor. The loss of stability induced ripples in the load, PAC induced erratic charge discharge cycles of the ESS, and reduced reliability of the entire microgrid ecosystem. Because the Kalman filter solves (a stochastic) control problem by approximation, this unique approach allows for the tracking and diligent mitigation of any output power ripples being fed the load, not only increased energy capture, but significantly increasing the stability of the entire microgrid ecosystem performance.

II. LVDC Microgrid Architecture and Dynamic Modeling

A robust and accurate system model is fundamental for the design and validation of control strategies. This section outlines the configuration of the simulated LVDC microgrid and the mathematical models that describe the dynamic actions of their constituent components. The parameters and models are based on elaborate system design (that ensures that the analysis occurs within the context of a realistic application scenario).

A. System Configuration

The LVDC microgrid being investigated in this research, illustrated in Figure (a), is providing power to an industrial DC load. The main source of power is a PV array rated at 100 kW capacity. The output from the PV array is processed by a DC-DC boost converter which performs two functions: converts the variable PV voltage to a constant PV bus voltage of 380V as well as being the actuator of the MPPT algorithm. An Energy Storage System (ESS), consisting of a Lithium-ion battery bank with a high energy storage capacity, is connected to the DC bus through a bidirectional buck-boost converter. The ESS is important for reliability, providing buffering to the variability of the PV source and keeping the DC bus voltage steady. The primary load is a 5 HP Separately Excited DC Machine (SEM) which acts as a dynamic load to mimic an industrial application, [8] i.e. a rolling mill and/or machine tool, with speed control. The SEM is fed from the DC bus through a dedicated chopper circuit.

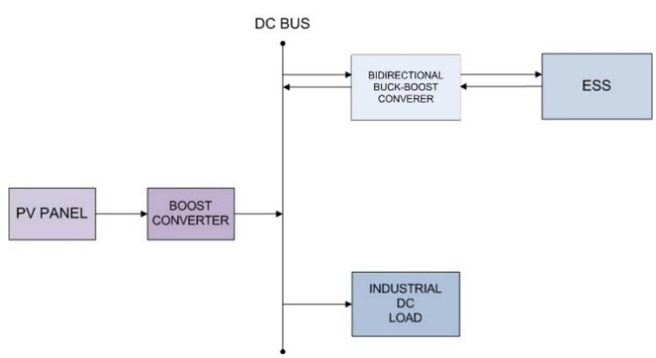

Fig. 1. Power Flow in Standalone LVDC Microgrid.

B. Photovoltaic (PV) Source Model

The single-diode equivalent circuit model is a well-known and accepted representation of the PV array's electrical behavior. This model models the nonlinear current-voltage (I-V) characteristics of the array at different irradiance and temperature conditions accurately. For PV array output current (I) Equation:

$$I = I_{ph} - I_o\left(e^{\frac{V+IR_s}{nV_t}} - 1\right) - \frac{V + IR_s}{R_{sh}}$$

Where I_{ph} is the photocurrent generated by solar irradiance, Io is the reverse saturation current of the diode, V and I are the terminal voltage and current of the array, respectively, Rs is the equivalent series resistance representing internal losses, R_{sh} is the shunt resistance representing leakage currents, n is the diode ideality factor, and V_t is the thermal voltage. This model forms the basis for simulating the power generation characteristics that the MPPT algorithms aim to optimize.

C. Power Conversion Stage: DC-DC Boost Converter

The DC-DC boost converter serves as the interface between the PV array and the DC bus. It is controlled through a high frequency switching signal and its duty cycle, D, is modulated by the MPPT controller. The ideal steady state voltage and current transfer functions of the boost converter are expressed as:

$$V_{out} = \frac{V_{in}}{(1-D)}$$

$$I_{out} = I_{in}(1-D)$$

where (V_{in}, I_{in}) are the input voltage and current from the PV array, and (V_{out}, I_{out}) are the output voltage and current delivered to the DC bus. The converter is modeled with its essential passive components, an inductor (L) and a capacitor (C). The size of the inductor and capacitor design can handle power flow through the converter; once sized, it can limit voltage and current ripple at the designated switching frequency of 5 kHz.

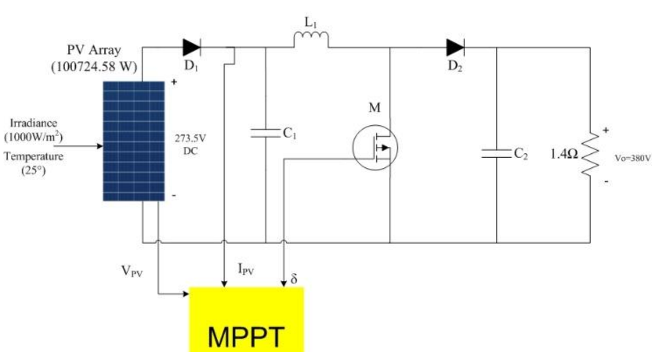

Fig. 2. MPPT with Boost Converter

D. Dynamic Load Model: Separately Excited DC Machine (SEM)

The SEM is a crucial element of the system because it acts as a dynamic load, and its performance is influenced by the quality of supplied power [8]. An SEM has complex electromechanical dynamics, unlike a resistive load, and the behavior of an SEM can be described with a few coupled differential equations.

Electrical dynamics are governed by the armature circuit equation:

$$V_t = I_a R_a + L_a \frac{di_a}{dt} + E_b$$

where Vt is the terminal voltage applied to the armature, I_a is the armature current, Ra and La are the armature resistance and inductance, and E_b is the back electromotive force (EMF), which is proportional to the motor's speed ($E_b = K_e \phi \omega_m$).

The mechanical dynamics are described by the torque balance equation:

$$J\frac{d\omega_m}{dt} = T_e - T_L - B\omega_m$$

where J is the moment of inertia of the rotor and connected load, ω_m is the angular speed of the rotor, T_e is the electromagnetic torque developed by the motor ($T_e = K_t \phi I_a$), TL is the mechanical load torque, and B is the viscous friction coefficient. The stable, ripple-free operation of the SEM relies upon stable well-regulated input power, which ties the mechanical performance of the SEM to the performance of the MPPT controller.

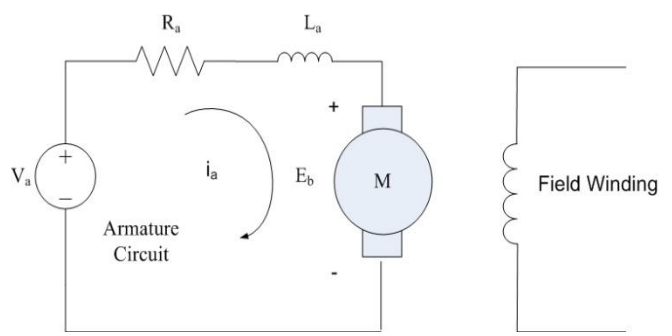

Fig. 5. Equivalent Circuit diagram for DC Motor

E. Energy Storage System (ESS) Integration

The ESS is represented as a Lithium-ion battery pack with a capacity of 1333.3 Ah, 300V nominal. Its main purpose is to absorb power from or inject power into the DC bus to balance generation and load and to stabilize the bus voltage. The ESS was modelled with an initial State of Charge (SOC) of 40% to provide adequate capability to both charge (when PV generation exceeds load) and discharge electrical energy (when load outweighs generation) [7]. Lastly, the bidirectional converter will allow charging and discharging to occur under strict control, which is essential for the effective management of this energy and the effective lifespan of the battery pack.

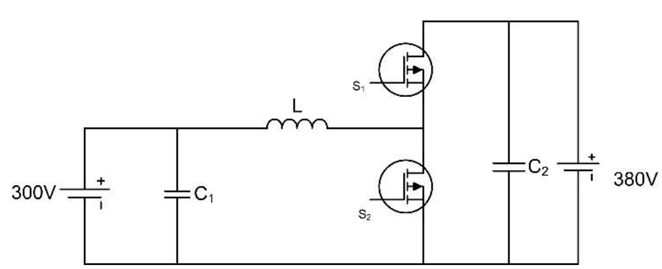

Fig. 4. Bidirectional Converter Connecting ESS to DC Bus

TABLE I
SYSTEM COMPONENT AND SIMULATION PARAMETER

Components	***Parameter***	***Values***
P.V Array	Maximum Power	100.7 kW
	MPPT Voltage	273.5 V
	Standard Temprature	25℃
DC-DC Boost Converter	Input Voltage Range	~273.5 V
	Output Voltage	380 V
	Switching Frequency	5 kHz
	Inductor	0.209 mH
	Outer Capacitor	3888 µF
Separately Excited DC Machine (SEM)	Rated Power	5 HP
	Rated Armature voltage	240 V
	Rated Speed	1750 RPM
	Armature Resistance	0.6 Ω
	Armature inductance	0.012 H
	Total Inertia	1 Kg-m^2
Energy Storage System	Input Voltage Range	Lithium Ion
	Nominal Voltage	300V
	Rated Capacity	1333.3 Ah
	Initial State of Charge	40%

III. MPPT CONTROL STRATEGY FORMULATION

This section contains a complete theoretical and mathematical analysis of the two MPPT control strategies examined in this report. The first is the traditional P&O algorithm with a PI controller and chosen to provide the performance base line. The second is the predictive controller which is proposed in this report and based on the Kalman filter.

A. Baseline Control: Perturb & Observe with PI Regulation

The P&O algorithm is an iterative, hill-climbing method that seeks the MPP by introducing small perturbations to the PV array's operating point and observing the effect on the output power. The logic, illustrated in the flowchart in Figure 2, is straightforward. At each control interval, the algorithm measures the PV voltage (V(k)) and current (I(k)) to calculate the power (P(k)). This power is compared to the power from the previous interval (P(k−1)). If the power has increased ($\Delta P>0$), the perturbation continues in the same direction. If the power has decreased ($\Delta P<0$), the direction of perturbation is reversed.

In this implementation, the P&O logic generates a reference voltage (V_{ref}). A Proportional-Integral (PI) controller is then employed in an inner control loop to drive the actual measured PV voltage (V_{pv}) towards this reference voltage by adjusting the duty cycle of the boost converter. The PI controller helps to smooth the response and regulate the voltage, but it cannot overcome the fundamental limitations of the P&O logic itself, namely the steady-state oscillations and slow dynamic response.

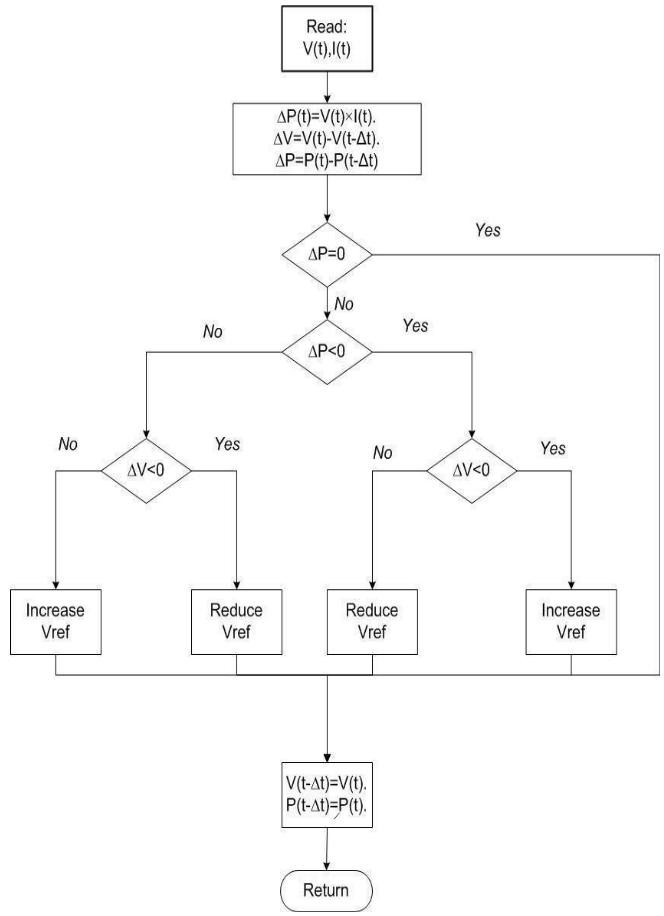

Fig. 5. P&O with PI Technique Flowchart.

B. Proposed Predictive Control: Kalman Filter-Based State Estimation for MPPT

The proposed approach goes beyond the reactive character of P&O and treats MPPT as a predictive state-estimation problem [4]. This conceptual leap is important. Instead of randomly perturbing the system, the Kalman filter will use the state-space model to predict the system's behavior and extract the noise from measurements. The result is more accurate and stable tracking of the MPP.

1. Framing MPPT as a State-Estimation Problem

The basic concept is to treat the PV system's operating point as a dynamic state variable, defining a solution-space that varies over time. The PV voltage and current measurements can be viewed as noisy observations of this true underlying state. The purpose of the Kalman filter is to recursively compute the optimal estimate of this state variable by using the model-based prediction and the measurements, which are noisy observations of the state. This allows the algorithm to be inherently robust to the sensor noise and rapid variations that may confuse simpler model methods such as P&O.

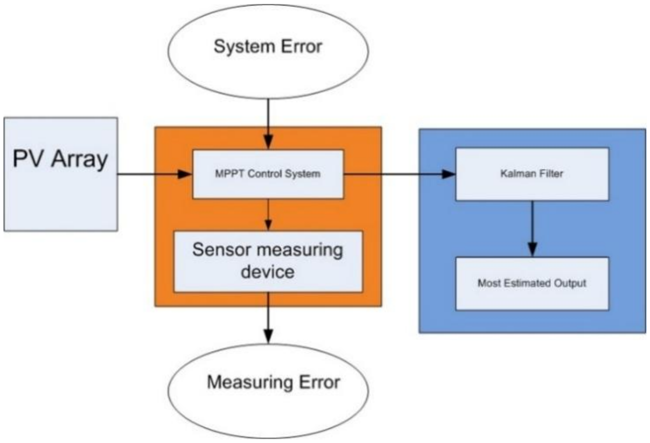

Fig. 6. Kalman Filter-Based MPPT Error Correction Architecture.

2. System State-Space Model

The Kalman filtering approach can be applied only if the system can be described in discrete-time state-space format. A standard way to formulate MPPT is to identify the state vector, xk, in terms of PV voltage and the slope of the P V curve. The MPP is achieved when the slope of the P-V curve is equal to zero. Thus, the general state-space model is given by: State Transition Equation:

$$x_k = Ax_{k-1} + Bu_{k-1} + w_{k-1}$$

Measurement equation:

$$z_k = Cx_k + v_k$$

Here, xk is the state vector at time step k, uk is the control input (related to the duty cycle perturbation), and zk is the measurement vector (containing measured voltage and current). The Matrices A, B, and C define the system dynamics. The terms wk and vk are crucial; they represent the process of noise and measurement noise, respectively. They are assumed to be zero-mean, white Gaussian noise with covariance matrices Q and R. The Kalman filter is specifically designed to handle these stochastic uncertainties. The tuning of the covariance matrices, Q and R is critical to the workings of the filter. These matrices can be thought of as a measurable "trust" model. The real measurement noise covariance, R, represents our trust in the sensor data. If R has a large value, then you are saying that you trust the measurements are noisy, and so the filter is more inclined to depend on what it predicted from its internal model. The process noise covariance, Q, represents our trust in the accuracy of the system model. If Q has a large value, then you are saying that the system might be having some changes that are not completely captured by the model (for example, the irradiance suddenly changed very quickly), or allow the filter to depend more heavily on more recent measurements. The beauty of this tuning approach is that the user can easily tune the filter behavior to fit their own applications using the characteristics of the system and the conditions of the environment.

3. The Recursive Kalman Filter Algorithm

The Kalman filter operates in a two-step, recursive loop: prediction and correction (or update). This process is computationally efficient and well-suited for real-time implementation.

(a) **Prediction (Time Update):**

In this step, the filter uses the system model to project the state and its uncertainty (error covariance) from the previous time step (k−1) to the current time step (k). This yields a priori (before measurement) estimate.

- Project the state ahead:

$$x_{k|k-1} = Ax_{k-1|k-1} + Bu_{k-1}$$

- Project the error covariance ahead:

$$P_{k|k-1} = AP_{k-1|k-1}A^T + Q$$

(b) **Correction (Measurement Update):**

In this step, the filter uses the current measurement (zk) to refine a priori estimate, resulting in an improved a posteriori (after measurement) estimate.

- Compute the Kalman Gain (Kk): This gain determines how much weight is given to the new measurement versus the prediction. It is calculated as:

$$K_k = P_{k|k-1}C^T(CP_{k|k-1}C^T + R)^{-1}$$

- Update the state estimate with the measurement:

$$x_{k|k} = x_{k|k-1} + K_k\left(z_k - Cx_{k|k-1}\right)$$

- Update the error covariance:

$$P_{k|k} = (I - K_kC)P_{k|k-1}$$

4. Duty Cycle Generation

The eventual outcome of this recursion is Xk|k, the optimal estimate of the state of the system, which, from the state vector, provides an good estimate of the MPP voltage Vmpp. This estimate ultimately will be the setpoint for a PI or similar controller to provide the exact duty cycle that needs to be applied to the boost converter to drive the PV array to the MPP. What we achieve here is a closed loop system that tracks the true MPP with full and smooth tracking - even in noise and disturbances.

IV. COMPARATIVE PERFORMANCE EVALUATION AND DISCUSSION

This segment presents an extensive comparative evaluation of the MPPT controllers based on the P&O-PI and Kalman filter. The comparison is built upon a rigorous simulation framework that exercises the MPPT controllers under difficult, dynamic conditions that are representative of possible operational scenarios.

A. Simulation Framework and Test Conditions

The whole LVDC microgrid system consisting of the PV array, power converters, SEM load, and control algorithms were modeled and simulated with MATLAB/Simulink. This software package is able to accurately model the system's nonlinear dynamics and the high frequency switching characteristics of power electronics. To rigorously evaluate the dynamic response and stability of the controllers, a specific test scenario was proposed. Specifically, the system is subjected to a large step change in solar irradiance simulating the classic example of heavy clouds moving into the PV array. In this case, the irradiance starts a constant 1000 W/m², then drops through a step change to 250 W/m² at time t=1s and remains constant for one second and finally goes through another step change at t=2s back to 1000 W/m². The ambient temperature is held constant at 25 °C throughout the simulation. This transient condition was selected specifically to analyze tracking speed, stability and robustness for the different control algorithms.

B. Dynamic Response Under Irradiance Transients

The response of the key electrical parameters of the PV system: power, voltage, and current, will provide the clearest measurement of MPPT performance. Here we present, in figure(g), the comparative outcomes for the two controllers under the defined irradiance transient.

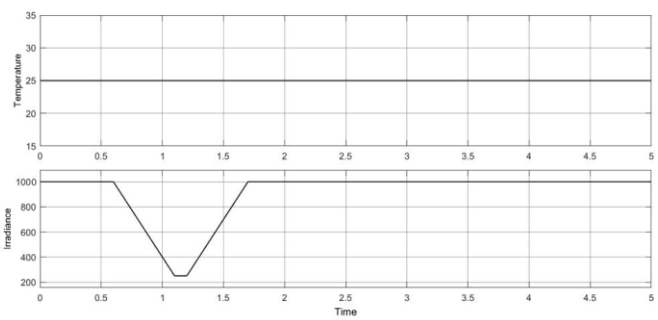

Fig. 7. Temperature and Irradiance Constraints

The P&O-PI, as a controller, creates oscillations in the PV power output, even during the first steady-state condition at 1000 W/m². From the trends seen in the previous graphs, it appears the controller reacts with a slow and uncertain response when the irradiance reduces, which results in extended hunting as it approaches the new lower MPP. When the irradiance restores to its high value at t = 2 s, the response again takes a longer time to settle, approximately 1.2 seconds. Furthermore, the PV voltage demonstrates a significant amount of ripples in the simulation, suggesting that the algorithm continues to perturb in an inefficient manner around the MPP.
In comparison, the Kalman filter controller produced a significantly improved response. The PV power curve is smooth, stable, and does not exhibit appreciable oscillation in the steady state. When the irradiance reduces and when the irradiance is restored, the controller reacts almost instantaneously, converges much quicker to the new MPP, and has less overshoot and undershoot. The settling time is much reduced, settlements within approximately 0.4 seconds. The PV voltage is remarkably stable in the last case, and this reflects the accuracy in estimating the optimal operating point through the filter, and without the perturbation.

C. Impact on Dynamic Load Stability

The ultimate test of the application of the MPPT controller is the effect upon the mechanical stability of the SEM load as the characteristics of the electrical power to the motor have a direct effect on its speed and torque characteristics. Figure (h) and Figure (i) show the Load-Power Stability of two Techniques below.
There is a severe instability arising from the P&O-PI controller resulting in the motor's performance. In an allowable range, the motor speed has a consistent power oscillation allowing for a ripple with a variation of ±1.5% around setpoint conditions during steady-state operation. During an irradiance transient the motor speed experiences a significant dip in speed of 100-150 RPM proportional to the irradiance profile; further by analogy to the electrical system signal response, the recovery by the P&O-PI controller is poor to transition back to steady-state conditions in any reasonable time. This is the performance of the motor as a load driven by the remaining components in the electrical system (battery, BMS, and controller). This performance would be unacceptable for many industrial applications where speed regulation is critical including textile mills, machine tools, etc [10].
For the Kalman filter based controller, the supplied power is smooth and stable mechanism for isolating the load can be achieved. The steady-state motor speed is approximately 1750 RPM controlled with a ripple of only ±0.2%. The same transient rate was supplied to the motor with a maximum speed drop of around 50 RPM; the recovery is fast, well damped to the irradiance profile and all components seem to be monitoring well enough to dynamically take in the change in electrical and mechanical and overall impact upon system response as the Kalman Filter tuned to these parameters really improved performance irregardless. Not only does this powerful result demonstrate the benefit at the system level of more advanced control to the electrical controller performance correlates to the improved moving performance and equipment reliability, highly obvious.

D. Quantitative Analysis of Tracking Efficiency and Power Quality

For a comparison of performance, the equivalent performance between the two controllers is encapsulated in Table II. The tracking efficiencies are presented as the ratio between the actual power extracted and the theoretical maximum power, permitting an evaluation of the considerable energy savings when utilizing the advanced controller. The P&O-PI controller demonstrates an efficiency of 92%, with the 8% loss mainly caused by the remaining steady-state oscillations in the steady-state and the slow dynamic tracking of the P&O-PI controller. The Kalman filter's design, which eliminates both loss mechanisms, brings the tracked efficiency to a much higher 98.5%. [6]
The following improvement in both efficiency and quality of power will also provide advantages to the whole microgrid. The smooth power delivery of the Kalman filter controller

would significantly reduce the duty cycle of the DC bus systems, especially the ESS. Specifically, with the erratic nature of power output from the P&O controller, the ESS would constantly be charging and discharging at high frequency duty cycles, essentially weathering the fluctuations in output power, which is not an effective solution and can speed up degradation of the battery system. With the stable power flow of the system because of the Kalman filter, they may manage their energies in a more effective way that maximizes control of the BCESS batteries, thus improving the lives of their batteries and the microgrid.

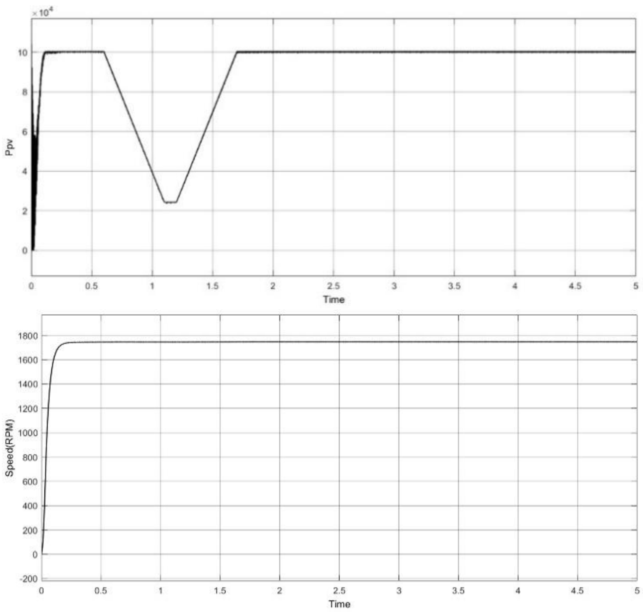

Fig. 8. Results for P&O based MPPT

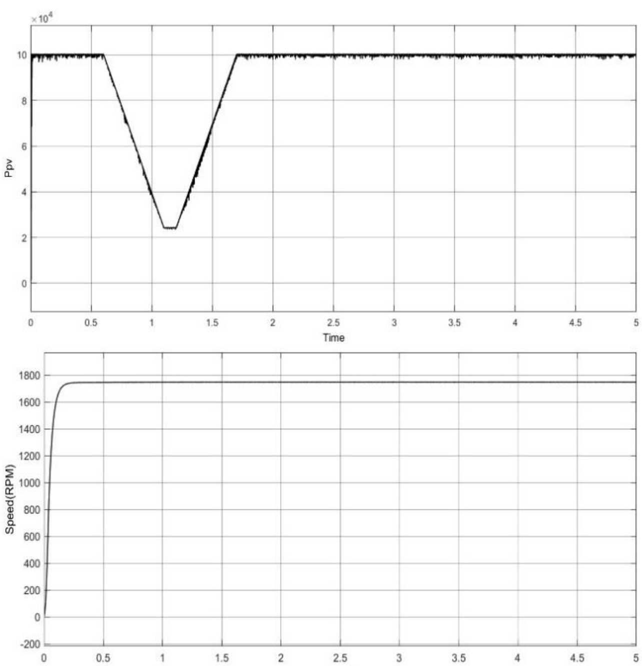

Fig. 9. Results for Kalman based MPPT

TABLE II
COMPARATIVE PERFORMANCE METRICS

Performance Metric	P&O-PI Controller	Kalman Filter MPPT	Improvement
Tracking Efficiency	92.0%	98.5%	+6.5%
Settling Time (after transient)	~1.2s	~0.4s	67% faster
Steady-State Power Ripple	Noticeable	Negligible (~0 W)	~100% reduction
SEM Speed Ripple	±1.5%	±0.2%	87% reduction

V. CONCLUSION AND FUTURE WORK

This paper undertook an equivalent design and comparative evaluation of a classic P&O-PI controller with a predictive Kalman filter-based MPPT algorithm for an LVDC microgrid supplying a dynamic motor load. The findings clearly illustrate the significant advantages of the predictive state-estimation approach that is being investigated in this paper as compared to the traditional reactive approach.

There are three main takeaways from this research:

1. **Improved Energy Capture and Efficiency:** The Kalman Filter-Based MPPT Tracking efficiency was 98.5%, significantly better than the P&O-PI controller tracking efficiency of 92%. This is because it virtually eliminated steady-state power oscillations and provided a dynamic response time to harsh irradiance conditions that were 67% faster than P&O-PI.
2. **Improved Mechanical Stability:** The research established a relationship between the power quality of the electrical power and the performance of the mechanical load. Provided ripple-free stable power, the Kalman Filter Controller provided a better averaging which resulted in an 87% decrease in speed ripple for the Separately Excited DC Machine which is highly valuable in precision industrial applications.
3. **System level benefits:** The Kalman Filter enhanced mechanical stability and power quality had cascading effects throughout the microgrid, likely leading to improved performance of the DC bus components, and improved stability and efficiency performance of the Energy Storage System - all improving overall system reliability and longevity.

In conclusion, for modern LVDC microgrids where performance, stability, and reliability are crucial - predictive state-estimation approaches like the Kalman filter, are not just an incremental improvement, but a enabling technology. Response capabilities of control are essential to realize the opportunities renewable energy sources have to offer in complex, unstructured conditions.

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